Predicting Coronary Heart Disease from BRFSS Survey Data

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***Abstract*—Early identification of individuals at risk of coronary heart disease is a valuable public-health goal. We build a binary classifier over the 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey, predicting whether a respondent reports a diagnosed heart attack from 321 self-reported health and lifestyle questions. We implement six linear methods from scratch in NumPy, design a preprocessing pipeline (one-hot encoding, imputation, clipping, standardization), and select the decision threshold by maximizing the F1 score on out-of-fold predictions—a step made necessary by the 8.8% positive class. The best model, regularized logistic regression trained on all features, reaches a five-fold cross-validated F1 score of 0.4290, nearly doubling the F1 obtained with the default threshold.**

I. Introduction

The task is a supervised binary classification problem: given the answers of a respondent to the 2015 BRFSS survey [1], predict the `MICHD` variable, which records whether the respondent has been diagnosed with coronary heart disease (myocardial infarction). The training set contains 328,135 samples described by 321 raw features and the test set 109,379 samples. Only 8.83% of the training labels are positive, so accuracy is misleading—a trivial all-negative classifier scores 91%—and the task is evaluated with the F1 score, which balances precision and recall on the positive class.

Our contributions are (i) a NumPy-only implementation of six linear models, (ii) a preprocessing pipeline that turns raw survey answers into a standardized design matrix, and (iii) a threshold-selection procedure that maximizes F1 on out-of-fold predictions. We validate every choice with stratified five-fold cross-validation and report an ablation of the most impactful decisions.

II. Data and Preprocessing

The BRFSS 2015 data [1] mixes continuous answers (BMI, age) and categorical answers (smoking status, income bracket) with survey metadata (weights, dates, sequence numbers). We drop all metadata and any column with more than 80% missing values. BRFSS sentinel codes are decoded: values such as 88, 888 and 555 ("no such value") and 77, 99, 777 and 999 ("don't know"/"refused") are treated as missing. Columns with at most 15 distinct values are one-hot encoded (most frequent level as reference, rare categories merged into it, missing as its own indicator); continuous features are imputed with the training median, receive a missingness indicator, and are clipped to the 0.1%–99.9% training quantiles. Columns that are linear combinations of others are removed, all features are standardized to zero mean and unit variance using training statistics, and a bias column is appended. Labels are mapped from $\{-1, +1\}$ to $\{0, 1\}$. The full pipeline yields 371 features.

As an ablation, we also consider a hand-picked set of 21 interpretable features (64 columns after one-hot encoding) motivated by the diabetes risk factors identified by Xie et al. [2] and curated for this heart-disease task by Teboul [3].

III. Methods

We implement the six linear methods from scratch using only NumPy: least squares and ridge regression via the normal equations, linear regression with gradient descent and stochastic gradient descent, and logistic regression with and without ℓ_2 regularization, trained by gradient descent on the logistic loss. Hyperparameters—regularization strength, step size and iteration count—are selected on a small grid with five-fold cross-validation; the best configuration is regularized logistic regression with $\lambda = 10^{-4}$, step size $\gamma = 0.5$ and 500 iterations.

Because the positive class is rare, thresholding scores at zero (probability 0.5) is suboptimal. For every model we compute out-of-fold scores with stratified five-fold cross-validation (seed 1) and choose the threshold that maximizes F1 on those scores. The same folds are used for all models, so comparisons are paired.

IV. Results

The best model, regularized logistic regression ($\lambda = 10^{-4}$, $\gamma = 0.5$, 500 iterations) trained on all features, reaches a five-fold cross-validated F1 of 0.4290 (± 0.0042) with 0.8768 accuracy and an F1-optimal threshold of -1.3153.

Figure 1 shows F1 as a function of the decision threshold. The curve is strongly peaked: the naive threshold $t = 0$ gives F1 0.233, whereas the optimum $t \approx -1.315$—predicting the positive class more often—gives 0.4290. For imbalanced data the threshold is therefore a first-order hyperparameter.

Table I compares representative models trained on all features. All well-behaved linear models cluster in a narrow band (F1 0.419–0.429), consistent with a weak, largely linear signal; SGD is the exception, converging slowly and noisily. Ablations: (i) using all 371 features instead of the selected 64-column subset raises F1 from 0.4125 to 0.4290, the single largest gain; (ii) ℓ_2 regularization is nearly neutral (0.4288

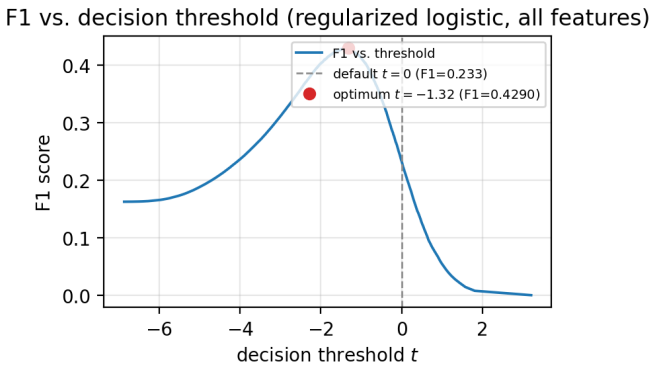


Fig. 1. F1 score as a function of the decision threshold t for the best model (regularized logistic regression on all features). The red dot marks the F1-optimal threshold ($t \approx -1.315$, F1 $= 0.4290$); the dashed line marks the naive threshold $t = 0$ (F1 $= 0.233$).

TABLE I
FIVE-FOLD CROSS-VALIDATED RESULTS OF REPRESENTATIVE MODELS TRAINED ON ALL FEATURES, WITH THE F1-OPTIMAL THRESHOLD APPLIED TO THE OUT-OF-FOLD SCORES. RIDGE USES λ=10^{-3}; LOGISTIC METHODS USE γ=0.5 AND 500 ITERATIONS; REGULARIZED LOGISTIC ADDS λ=10^{-4}.

Model	F1	Accuracy	Threshold
Least squares	0.4197	0.8677	0.219
Ridge	0.4199	0.8644	0.214
Linear reg. (GD)	0.4187	0.8730	0.225
Linear reg. (SGD)	0.3168	0.8252	0.254
Logistic reg.	0.4288	0.8736	-1.361
Reg. logistic	**0.4290**	**0.8768**	-1.315

to 0.4290) but stabilizes the loss; and (iii) optimizing the threshold raises F1 from 0.233 to 0.4290.

V. DISCUSSION AND CONCLUSION

Regularized logistic regression on all features with an F1-optimized threshold reaches a five-fold cross-validated F1 of 0.4290 on a noisy, self-reported survey where the linear baselines lie close to one another. The analysis shows that feature breadth and threshold selection matter more than the choice among linear models, consistent with the advice that data and evaluation often dominate the learner [4]. All results are reproducible: `python run.py` retrains the reported model and writes the AIcrowd submission. A limitation is that linear models cannot capture feature interactions; in the follow-up project we plan to explore non-linear and ensemble methods.

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